Minimization of Loss function to obtain BEST model parameters

$$Loss = \frac{1}{2}(Y - \hat{Y})^T(Y - \hat{Y}) = \frac{1}{2}(Y - XW)^T(Y - XW)$$

$$Loss = \frac{1}{2}[Y^TY + (XW)^T(XW) - Y^T(XW) - (XW)^TY]$$

$$Loss = \frac{1}{2}[Y^TY + (XW)^T(XW) - 2(XW)^TY]$$

Remember

X is a matrix, W is a vector,

XW is a vector

$$Y^T(XW) = (XW)^TY$$

$\frac{\partial Loss}{\partial W} = 0$ Get Best **W** that minimizes the loss

$$\frac{\partial Loss}{\partial W} = 0 = \frac{1}{2}[(XW)^T(X) + (X)^T(XW) - 2(X)^TY]$$

Remember

$$X^T(XW) = (XW)^TX$$

$$\frac{\partial Loss}{\partial W} = 0 = \frac{1}{2}[2X^TXW - 2X^TY] \rightarrow (X^TX)\underline{W} = X^TY \rightarrow$$

$$W = (X^TX)^{-1}X^TY$$